

Khwaja Yunus Ali University

Founder: Dr. M. M. Amjad Hussain

School of Science & Engineering

Department of CSE & EEE

Spring - 2024

Course Outline: Physics Lab

Part – A:

1. **Course Code:** CSE 0533 1102 (CSE) & PHY 0533 1202 (EEE)
2. **Course Title:** Physics Lab
3. **Course Type:** Related
4. **Level:** 1.1 (1st Year 1st Semester for CSE) & 1.2 ((1st Year 2nd Semester for EEE)
5. **Academic Session:** Spring - 2024
6. **Course Instructor:** Md. Matiur Rahman
7. **Prerequisites:** N/A
8. **Credit Value:** 1.5
9. **Contact hours/week:** 3.00 hours
10. **Total Marks:** 100
11. **Rationale of the Course:** This course is to introduce one to the proper methods for conducting controlled physics experiments, including the acquisition, analysis and physical interpretation of data. The course involves experiments which illustrate the principles of electricity, heat and thermodynamics, waves and oscillations and optics.

12. Course Objectives

By the end of the three-course intro lab sequence, students should be able to:

- Collect data and revise an experimental procedure iteratively and reflectively,
- Evaluate the process and outcomes of an experiment quantitatively and qualitatively,
- Extend the scope of an investigation whether or not results come out as expected,
- Communicate the process and outcomes of an experiment, and
- Conduct an experiment collaboratively and ethically.

13. Course Learning Outcomes (CLOs) and its mapping with PLOs:

CLOs	After successful completing this course, students will be able to.	PLO Mapping
CLO1	To enhance students' knowledge in experimental physics for higher study and research activities.	PLO1, PLO4 & PLO12
CLO2	To enable students to operate the instruments of Physics.	PLO4 & PLO5
CLO3	To provide an opportunity to data processing and analysis of the different experiments regarding physics.	PLO1, PLO2 & PLO4
CLO4	To develop the skills about the report design, writing and handling related to the lab works.	PLO2, PLO3 & PLO4
CLO5	To make students capable of using the ideas of Physics course to perform experiments.	PLO7, PLO8 & PLO11

Part – B:**14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:**

Week	Topic	Teaching Strategy	Assessment Technique	Alignment with CLOs
01	Introduction + Discussion + Precaution	Interactive discussion / Group activity / Discussion / Presentation / Laboratory Work	Quiz / Class test / Assignment / Presentation / Midterm / Final exam.	CLO1, CLO2, CLO3, CLO4 & CLO5
02	Determination of the focal length and hence the power of the concave lens by using an auxiliary convex lens.			
03	Determination of the thickness of a glass plate with a Spherometer.			
04	Determination of the refractive index of a liquid by pin method using a plane mirror and a convex lens.			
05	Continuous Assessment + Theoretical Discussion + Determination of the refractive index of a liquid by pin method using a plane mirror and a convex lens.			
06	Continuous Assessment + Theoretical Discussion			
07	Determination of the spring constant and effective mass of a given spiral spring.			
08	Determination of the value of 'g' by means of a compound pendulum.			
09	Continuous Assessment + Theoretical Discussion			
10	Determination of the specific resistances of a wire using a Meter bridge.			
11	Determination of the resistance of a Galvanometer by half-deflection method.			
12	Determination of the unknown resistances and to verify the series – parallel resistance by means of a Post Office Box.			
13	Continuous Assessment + Theoretical Discussion + Determination of the moment of inertia of a fly wheel about its axis of rotation.			
14	Continuous Assessment + Theoretical Discussion + Determination of the surface tension of water by capillary tube method and hence to verify Jurin's Law.			

Part – C:**15. Assessment and Evaluation****1. Assessment and Evaluation with marks distributions****1. Assessment Strategy**

Students are evaluated out of total 100 Marks. There are different types of assessment tools. In **the continues assessment**, the total marks is 50. In the **Final Examination**, total marks is 50 and allotted time is **2 Hours**. The detail with **CLOs-Assessment Mapping** is given below.

CLOs	Total (100%)						
	Continues assessment (50%)		Final Exam (50%)				
	Attendance (10%)	Lab Report (40%)	Attendance (5%)	Behavior & Participations (5%)	Theoretical Concept (10%)	Viva-voce (10%)	Written Exam (20%)
CLO1	√	√	√	√	√		√
CLO2	√		√		√		√
CLO3	√	√	√				√
CLO4	√	√	√				√
CLO5	√		√	√	√		√

2. Evaluation System (Grading System)

Numerical Number	Letter Grade		Grade Point
80 % and above	A+	(A Plus)	4.00
75 % to less than 80 %	A	(A Regular)	3.75
70 % to less than 75 %	A-	(A Minus)	3.50
65 % to less than 70 %	B+	(B Plus)	3.25
60 % to less than 65 %	B	(B Regular)	3.00
55 % to less than 60 %	B-	(B Minus)	2.75
50 % to less than 55 %	C+	(C Plus)	2.50
45 % to less than 50 %	C	(C Regular)	2.25
40 % to less than 45 %	D		2.00
Less than 40 %	F		0.00

Part – D:**16. Learning Materials:****1) Recommended Reading**

- Giasuddin Ahmad and Md Shahabudin - Practical physics - Last Edition
- Jerry D. Wilson - Physics Laboratory Experiments - Last Edition
- Matthew French - Physics Lab Experiments - Last Edition

2) Supplementary Reading

- Web Source Link
- Journal papers

3) Others:

- Lecture Outlines
- PowerPoint File
- Webpage